

The bootstrap

The bootstrap

- ▶ *Example:* Invest fraction α of a sum in an asset with return X and $1 - \alpha$ of it in an asset with return Y . Two risky assets.
- ▶ The return will be

$$R = \alpha X + (1 - \alpha)Y$$

- ▶ Choose α so that the variance of the return of the portfolio

$$\text{Var}(R) = \alpha^2 \sigma_X^2 + 2\alpha(1 - \alpha)\sigma_{XY} + (1 - \alpha)^2 \sigma_Y^2$$

is minimized.

- ▶ $\implies$

$$\alpha_{\min} = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - \sigma_{XY}}$$

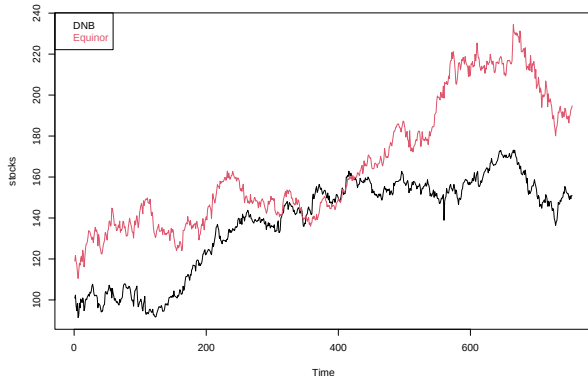
Sampling distribution for $\hat{\alpha}_{\min}$

$\alpha_{\min}$ estimated by $\hat{\alpha}_{\min}$ where σ_X^2 is estimated by the sample variance $\hat{\sigma}_X^2$, σ_Y^2 by the sample variance $\hat{\sigma}_Y^2$ and σ_{XY} by the sample covariance $\hat{\sigma}_{XY}$

$$\hat{\alpha}_{\min} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}$$

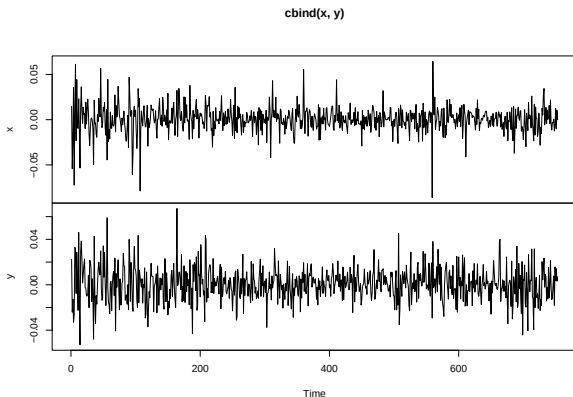
DNB and Equinor prices

```
load("stocks.RData")  
plot.ts(stocks,plot.type = "single",col=c(1,2))  
legend("topleft",c("DNB","Equinor"),text.col = c(1,2))
```



DNB and Equinor returns

```
x=diff(log(stocks$dnb))  
y=diff(log(stocks$equinor))  
plot.ts(cbind(x,y),plot.type = "multiple",col=c(1,2))
```



DNB and Equinor

```
vx=var(x)
```

```
vx
```

```
## [1] 0.0002186398
```

```
vy=var(y)
```

```
vy
```

```
## [1] 0.0002272927
```

```
cxy=cov(x,y)
```

```
cxy
```

```
## [1] 0.0001059944
```

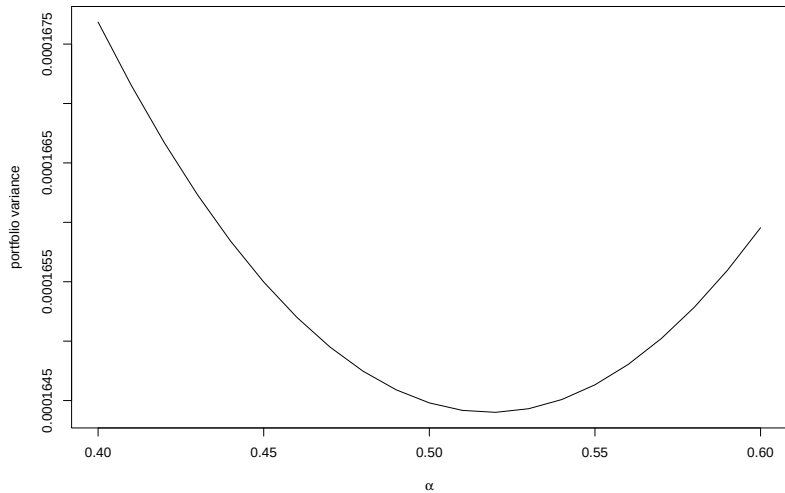
DNB and Equinor

```
alpha=(vy-cxy)/(vx+vy-2*cxy)
alpha
```

```
## [1] 0.5184935
```

- ▶ $\therefore$ The portfolio that minimizes the variance consists of 52% of DNB and the rest Equinor.

Variance as a function of $\hat{\alpha}$



A function to compute $\hat{\alpha}_{min}$

$$\hat{\alpha}_{min} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}$$

```
alpha.min=function(x,y)
{
  vx=var(x)
  vy=var(y)
  cxy=cov(x,y)
  amin=(vy-cxy)/(vx+vy-2*cxy)
  return(amin)
}
alpha.min(x,y) # TEST FUNCTION
```

```
## [1] 0.5184935
```

Bootstrapping $\hat{\alpha}_{min}$ to estimate the sampling distribution

- ▶ Data set: $(x_1, y_1), \dots, (x_n, y_n)$ (For our data, $n = 753$)
- 1. Draw n pairs from the sample with replacement. $(x_i^{(1)}, (y_i^{(1)}), i = 1, \dots, n$, is the first bootstrap sample.
- 2. Compute $\hat{\alpha}^{(1)}$
- 3. Repeat 1-2 B times to obtain $\hat{\alpha}^{(1)}, \hat{\alpha}^{(2)}, \dots, \hat{\alpha}^{(B)} = \hat{\alpha}$
- ▶ It can be shown that $\hat{\alpha}$ is a draw from the sampling distribution of $\hat{\alpha}_{min}$. We can e.g. estimate $Var(\hat{\alpha})$ by

$$\widehat{Var}(\hat{\alpha}) = \frac{1}{B-1} \sum_{b=1}^B (\hat{\alpha}^{(b)} - \bar{\hat{\alpha}})^2$$

Step 1 in R

Data set: $(x_1, y_1), \dots, (x_n, y_n)$ (For our data, $n = 753$)

1. Draw n pairs from the sample with replacement. $(x_i^{(1)}, (y_i^{(1)}),$
 $i = 1, \dots, n$, is the first bootstrap sample.

```
head(x,5)
```

```
## [1]  0.01475679 -0.05417573 -0.02294157  0.03574584 -0.07226977
```

```
n=length(x)
```

```
indboot=sample(1:n,size = n,replace = TRUE)
```

```
xboot=x[indboot]
```

```
yboot=y[indboot]
```

Step 2 in R

2. Compute $\hat{\alpha}^{(1)}$

```
head(x,5)
```

```
## [1]  0.01475679 -0.05417573 -0.02294157  0.03574584 -0.07226977
```

```
head(xboot,5)
```

```
## [1]  0.011718884 -0.008406128 -0.003773589  0.044537854  0.005904076
```

```
a1=alpha.min(xboot,yboot)
```

```
a1
```

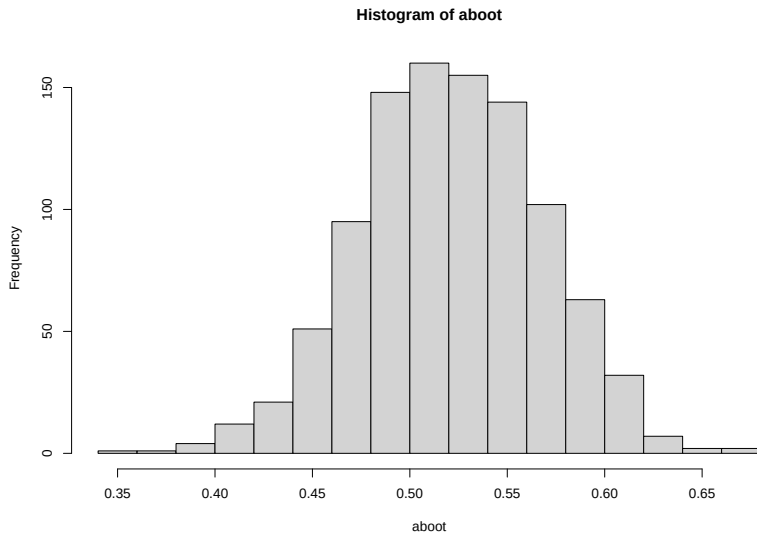
```
## [1] 0.4645268
```

Step 3 in R

3. Repeat 1-2 B times to obtain $\hat{\alpha}^{(1)}, \hat{\alpha}^{(2)}, \dots, \hat{\alpha}^{(B)} = \hat{\alpha}$

```
B=1000
aboot=matrix(0,B,1)
for(b in 1:B)
{
  indboot=sample(1:n,size = n,replace = TRUE)
  xboot=x[indboot]
  yboot=y[indboot]
  aboot[b]=alpha.min(xboot,yboot)
}
hist(aboot)
```

The estimated sampling distribution



A 95% confidence interval for the optimal α

```
quantile(about, probs=c(0.025, 0.975))
```

```
##          2.5%          97.5%
```

```
## 0.4295908 0.6066036
```

If we computed such intervals for 100 different samples we would expect that 95 of them would cover the true (but unknown) α_{min} .

So how do we know that this is really a good estimator of the sampling distribution?

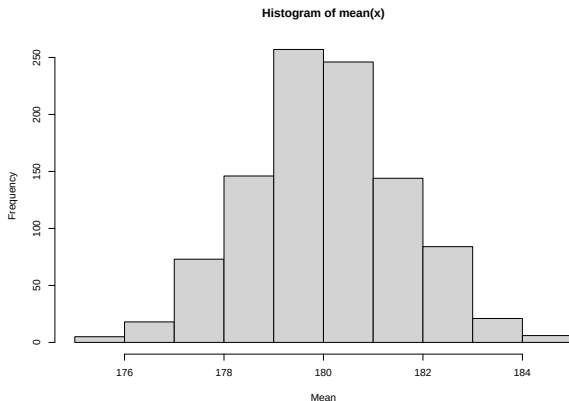
- ▶ Let us go back to a familiar example from last lecture
- ▶ X = height of a randomly chosen Norwegian adult
- ▶ Want to estimate the expected height, $\mu = E(X)$
- ▶ Draw a sample of $n = 100$, $x_1, x_2, \dots, x_n$
- ▶ Estimate $\hat{\mu} = \bar{x}$ (one single value)
- ▶ The sampling distribution of $\hat{\mu}$ is the distribution (histogram) of the $\bar{x}$ if we repeatedly sampled 100 observations.
- ▶ We know that it is well approximated by $N\left(\mu, \frac{\sigma^2}{n}\right)$ for large samples.
- ▶ Thus, the bootstrap estimation of the sample distribution should look approximately like $N\left(\mu, \frac{\sigma^2}{n}\right)$

1000 different samples of 100 observations each

```
# xpop a population of 4 million with mean 180 and sd 15
head(xpop)
n=100
nrepl=1000
res=matrix(0,nrepl,1)
for(irepl in 1:nrepl)
{
  x=sample(xpop,size = n)
  res[irepl]=mean(x)
}
hist(res,xlab="Mean",main="Histogram of mean(x)")
```

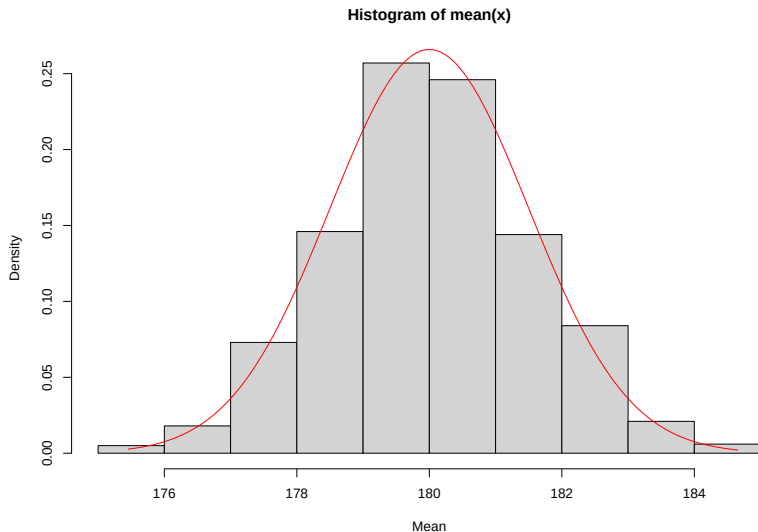
1000 different samples of 100 observations each

► $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{1000}$



With the large sample approximation

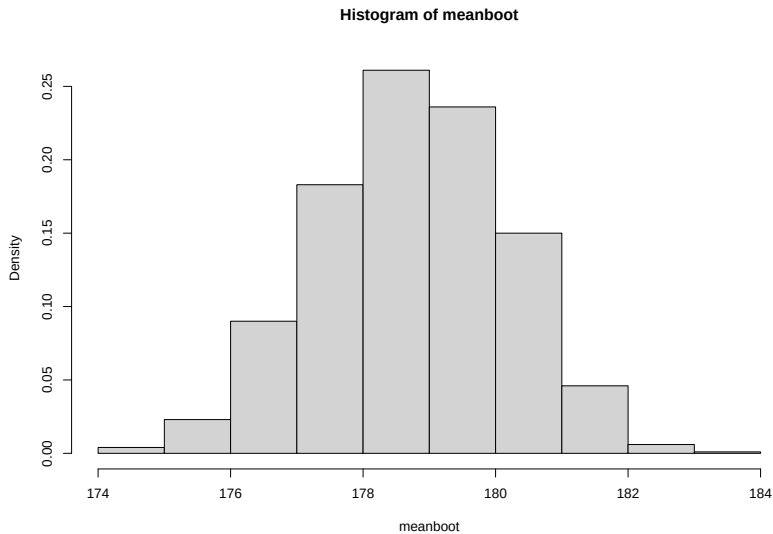
$$\therefore \bar{X} \sim N(180, 2.25)$$



Bootstrapping $\bar{X}$

```
n=100
x=sample(xpop,size=n) # Sample from population
B=1000
meanboot=matrix(0,B,1)
for(b in 1:B)
{
  indboot=sample(1:n,size = n,replace = TRUE)
  xboot=x[indboot] # Sample from sample
  meanboot[b]=mean(x)
}
hist(meanboot)
```

Bootstrapping $\bar{X}$



Confidence intervals

```
m=mean(x)
s=sd(x)
# Large sample approximation
c(m-1.96*sd(x)/sqrt(n),m+1.96*sd(x)/sqrt(n))
```

```
## [1] 175.8247 181.5648
```

```
# Bootstrap
quantile(meanboot,probs=c(0.025,0.975))
```

```
##      2.5%      97.5%
```

```
## 175.9320 181.3913
```